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Cobalt-Catalyzed Electrophilic Cyanation of Arylzinc Halides with *N*-Cyano-*N*-phenyl-*p*-methylbenzenesulfonamide (NCTS)

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Abstract: The cobalt-catalyzed cross-coupling of organozinc bromides with *N*-cyano-*N*-phenyl-*p*-methylbenzenesulfonamide (NCTS) is described. The same cobalt catalyst, cobalt(II) bromide, was used for both the synthesis of the organozinc species and the cross-coupling reaction. However in this case, a catalytic amount of zinc dust is necessary in the second step to release the low-valent cobalt. Under these mild conditions, moderate to excellent yields of different benzonitriles were obtained.

Keywords: benzonitriles; cobalt; cross-coupling; electrophilic cyanation; organozinc compounds

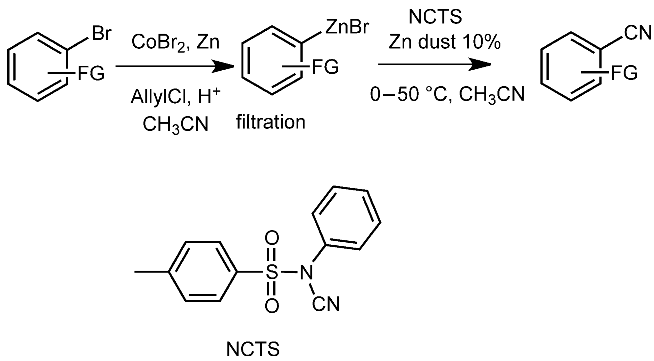
Benzonitriles play an important role in natural products, pharmacy, dyes and electronic materials.[1] Meanwhile the nitrile group allows a multitude of transformations to other important functional groups, such as amines, amidines, tetrazoles, aldehydes, amides.[2] In the early years, the Sandmeyer[3] and Rosenmund–von Braun[4] reactions were the two most employed methods for the introduction of a cyano group onto arenes using CuCN and relatively harsh conditions. However, both these drawbacks and the restricted functional tolerance, encourage researchers to find new methodologies to achieve the formation of most functionalized benzonitriles.

In the past few decades, transition metals (Pd, Cu, or Ni) catalyzed nucleophilic cyanation reactions of aryl substrates have emerged as powerful alternatives. Various cyanide sources, such as CuCN,[5] KCN,[6] NaCN,[7] $Zn(CN)_2$,[8] TMSCN,[9] acetone cyanohydrin[10] or $K_4[Fe(CN)_6]$[11] have been reacted with functionalized aryl halides to provide the corresponding aryl nitriles. However, some limitations remain: the high affinity of the cyanide ion towards the catalyst implies high catalyst loading, the toxicity of the cyanide reagents, and the generation of HCN. Then, alternative routes for the cyanation reactions employing “non-CN-unit” were described but they also generally require a stoichiometric amount of metal salt and harsh conditions.[12]

Electrophilic cyanations have been less studied compared to nucleophilic ones. However, they represent useful complementary alternatives, which sometimes overcome the above-mentioned drawbacks. Different aryl-organometallic reagents react with a variety of cyano sources. Aryllithium reagents react with cyanatobenzene[13] or ClCN at low temperature.[14] In the early years, the nucleophilic displacement of *p*-toluenesulfonyl cyanide[15] or 2-pyridyl cyanate[16] with phenylmagnesium bromide was also investigated. The scope of the Grignard reagents has only been explored in 2010 by Beller and co-workers who reported the electrophilic cyanation of aryl/heteroaryl Grignard reagents with *N*-cyanobenzimidazole.[17] In addition, the first Pd-catalyzed, Cu-mediated [CuTC: copper(I)-thiophene-2-carboxylate] efficient cyanation of arylboronic acids with benzyl thiocyanates was reported by Liebeskind and co-workers.[18] However, the main drawbacks of these methods are the preparation of the cyano reagent from cyanogen bromide, which is quite toxic and dangerous. To avoid this, later, Beller's group developed an elegant, novel and convenient synthesis of benzonitriles through aryl/heteroaryl Grignard reagents,[19] prepared *via* Knochel's procedure,[20] using a benign cyanating reagent, *N*-cyano-*N*-phenyl-*p*-methylbenzenesulfonamide (NCTS). NCTS is a bench-stable, easy to handle and environmentally benign electrophilic cyanating agent. Compared to the previous established electrophilic cyanating sources, its synthesis does not require the highly toxic cyanogen halides, or similar cyanating precursors.[21] Substituted arenes were cyanated in high yields and these methods tolerate both electronically rich/poor groups at any place of the arene.[19] This methodology is cost-effective and environmentally friendly. The same year, the same group extended this methodology to arylboronic acids using this time Rh catalysis.[22] More recently, new Rh,[23] Ru,[24] Cu,[25] or Co[26] catalyzed directed C–H cyanation reactions were developed using this user-friendly cyanation reagent. Nevertheless, the functionalization of unactivated C–H bonds required a directing group and high temperature. In the same manner, a direct oxidative cyanation of arenes under Fe(II) catalysis using an excess of aryl(cyano)-iodonium triflates as the cyanating agent has been successfully applied to a range of electron-donating substituents on aromatic substrates.[27] Although some elegant results were reported by C–H activation, organometallic reagents remain useful to functionalize aryl halides without a directing group. Nevertheless, organozinc compounds known for their broad functional tolerance were rarely employed in cyanation. Noteworthy, some twenty years ago, Knochel's group developed the only efficient transition metal-free procedure for the cyanation of a wide range of organozinc compounds at low temperature using toxic *p*-toluenesulfonyl cyanide.[28]

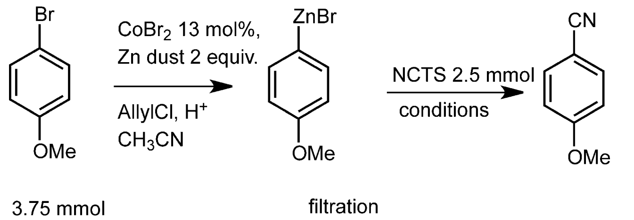
All these umpolung procedures have made important contributions to the synthesis of aryl nitriles. Many efficient methodologies have been developed. However, all these methods suffer from some drawbacks in that they require either the use of stoichiometric co-catalyst or additive, or expensive catalyst, or dangerous CN source. Moreover, high or very low temperatures are often necessary to ensure a good yield. Thus, despite impressive recent progress, the development of a mild, inexpensive, and simple procedure remains highly desirable. Arylzinc reagents, whose synthesis is well understood by our group, are good candidates as nucleophilic partners taking into account their special reactivity. Thus, this paper presents the cyanation of arylzinc reagents using the benign NCTS as the electrophilic CN source in order to form a variety of functionalized aryl nitriles (Scheme 1).

Using the method we developed to synthesize arylzinc bromides with a $CoBr_2$/Zn catalytic system, we first examined the reaction of *p*-methoxyphenylzinc bromide, formed by cobalt catalysis, with *N*-cyano-*N*-phenyl-*p*-toluenesulfonamide (Table 1). Adding the NCTS directly to the crude solution of arylzinc bromide affords the desired 4-methoxybenzonitrile in modest yield (Table 1, entry 1). Previously, we have shown that zinc can be easily removed,[29] but in this reaction, filtration decreases the yield (Table 1, entry 2). Carrying out the reaction at 50 °C instead of room temperature, increases the yield (Table 1, entry 3), whereas introduction of THF in the medium, inhibits the reaction (Table 1, entry 4). Moreover, addition of a catalytic amount of zinc dust in the second step is essential for the success of the cross-coupling



Scheme 1. Cobalt-catalyzed electrophilic cyanation of arylzinc species.

Table 1. Optimization of reaction conditions.

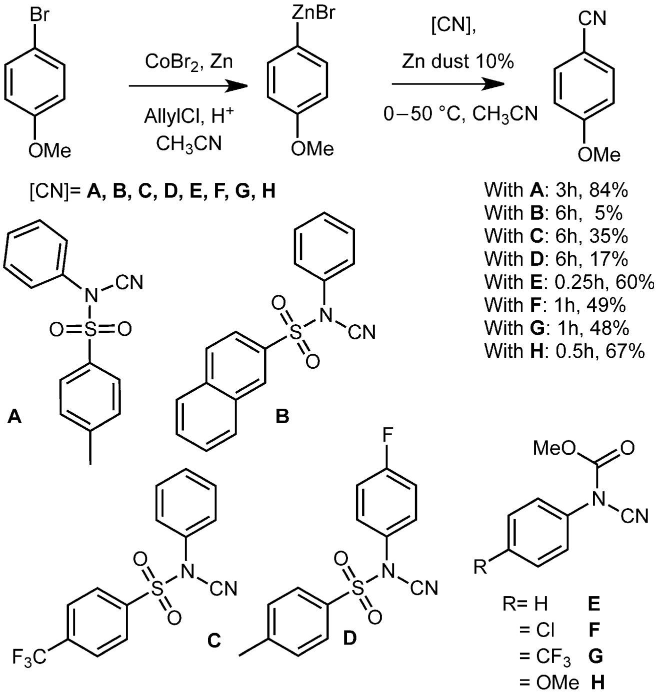


Entry	Conditions	Yield
1	0 °C to r.t., 3 h or 12 h	45%[a]
2	filtration of ArZnBr, 0 °C to r.t., 12 h	30%[b]
3	filtration of ArZnBr, 0 °C to 50 °C, 12 h	38%[b]
4	filtration of ArZnBr, add THF, 0 °C to 50 °C, 12 h	< 5%[b]
5	filtration of ArZnBr, add 10 mol% of Zn, 0 °C to r.t. 16 h	67%[b]
6	**filtration of ArZnBr, add 10 mol% of Zn, 0 °C to 50 °C 3**	**84%[a]**
7	filtration of ArZnBr, add 5 mol% of Zn, 0 °C to 50 °C 3 h	43%[b]
8	filtration of ArZnBr, add 20 mol% of Zn, 0 °C to 50 °C 3 h	50%[b]
9	filtration of ArZnBr,[c] 0 °C to 50 °C, 3 h	77%[b]

[a] Isolated yield.
[b] GC yield (decane as internal standard).
[c] ArZnBr formed with [$CoBr_2$(bpy)].

with NCTS when ArZnX is formed with the simple $CoBr_2$. Therefore, the yield increases on adding 10 mol% of Zn in the medium (Table 1, entry 5) and is better at 50 °C (Table 1, entry 6). This catalytic amount of zinc is optimum (Table 1, entries 6, 7, 8).

In order to understand the role of the catalytic amount of zinc dust in the second step, we have replaced it by 10 mol% of other metals such as Mg, Fe, Mn and Cu and obtained lower GC yields of, respectively, 31, 0, 53 and 14%. Replacement of Zn by a Lewis acid such as $ZnCl_2$ to facilitate the desired cyanation gave no cyanated product. Therefore, we thought that Zn dust should help the release of the



Scheme 2. Analogous cyanide sources.

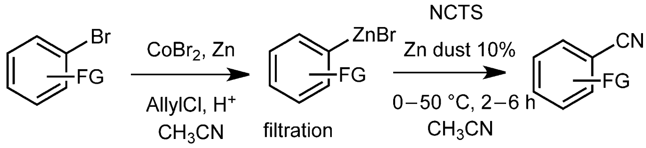
cobalt catalyst from the cyano group, since the electrophilic amination with RR′N-Cl carried out under the same conditions (acetonitrile) works well without zinc.[29a] This was further confirmed by preparing the arylzinc species with $[CoBr_2(bpy)]$ which is able to "protect" the cobalt from the cyano group (Table 1, entry 9). In this case, a good yield is obtained without supplementary addition of zinc in the second step. However, we have preferred to use commercially available $CoBr_2$ as catalyst and add zinc dust in the second step for economic reasons. This study shows also that cobalt is necessary for the cross-coupling of arylzinc derivatives with NCTS whereas it is not required for more reactive Grignard reagents.

Different substituted arylsulfonylcyanamides were tested (Scheme 2), which were synthesized by reacting substituted arylsulfonyl chlorides with substituted arylureas in pyridine at room temperature, to deliver **B**, **C** and **D**. No improvement was observed using these alternative cyano sources. Other N−CN reagents, *N*-carbomethoxy-*N*-arylcyanamides (**E**, **F**, **G**, **H**) (Scheme 2), were also synthesized in three steps from the corresponding commercial benzonitriles to see the influence of the groups on nitrogen and to compare them with arylsulfonylcyanamides. Compounds **E**, **F**, **G** and **H** were obtained in 26 to 58% overall yields. These cyano sources are much more reactive than arylsulfonylcyanamides and the reaction time can be reduced to 15 min. Good yields of cyanated products are obtained from **E** and **H** bearing an electron-donating substituent, whereas compounds **F** and **G** with an electron-withdrawing groups on the *N*-phenyl group are less efficient. However, these compounds are relatively unstable, so that the yield cannot be increased. Therefore, we focused our study on the NCTS previously used by Beller and we set out to explore the scope of the methodology.

A variety of arylzinc bromides bearing electron-donating or electron-withdrawing groups were coupled with NCTS under the optimized conditions defined above to afford functionalized benzonitriles in moderate to excellent yields (Table 2). It should be noted that some of them are quite difficult to obtain by the electrophilic cyanation of Grignard reagents, since ArMgX are too strong nucleophiles to tolerate some of those functional groups.

A methoxy group on the *para*-position of the arene is the more favorable substituent, the yields are lower when it is located at the *meta*- or *ortho*-position. (Table 2, entries 1 to 3). Small groups like methyl or fluoro on the *ortho*-position still provided excellent yields (Table 2, entries 4 and 5). Aryl halides without any substitutent (Table 2, entry 6), or with a variety of

Table 2. Cobalt-catalyzed cross-coupling of various functionalized arylzinc bromides with NCTS.



Entry	FG in FG-C_6H_4-CN[a]	Reaction time	Yield[b]
1	*p*-MeO (**1a**)	2.5 h	84%
2	*m*-MeO (**1b**)	2.5 h	57%
3	*o*-MeO (**1c**)	6 h	58%
4	*o*-Me (**1d**)	6 h	98%
5	2-F,5-Me (**1e**)	6 h	82%
6	H (**1f**)	6 h	76%
7	*p*-MeS (**1g**)	6 h	76%
8	*p*-MeOCO (**1h**)	6 h	40%
9	*p*-Cl (**1i**)	6 h	63%
10	*p*-F (**1j**)	6 h	74%
11	*p*-$MeSO_2$ (**1k**)	6 h	47%[c]
12	*p*-CF_3 (**1l**)	6 h	68%
13	*p*-Me_2N (**1m**)	6 h	68%
14	*p*-EtO_2C (**1n**)	6 h	72%
15	1,2-(methylenedioxy) (**1o**)	6 h	79%
16	3,5-diCF_3 (**1p**)	6 h	40%[d]
17	3,5-diMe (**1q**)	6 h	56%
18	1-naphthyl (**1r**)	6 h	79%
19	*p*-MeCO (**1s**)	6 h	34%[e]
20	*p*-PhCOPh (**1t**)	6 h	45%[e]
21	*p*-NC (**1u**)	6 h	50%[e]

[a] Ratio of ArBr to NCTS is 1.5:1.
[b] Isolated yield.
[c] Yield by ^{1}H NMR (*N*-phenyl-*p*-methylbenzenesulfonamide was isolated as well).
[d] Yield by GC (the corresponding ArH was also present).
[e] ArZnBr formed with $[CoBr_2(bpy)]$ instead of $CoBr_2$

functional groups, such as thioether, acetate, chloro, sulfone, fluoro, trifluoromethyl, dimethylamino, ester, dioxane provide moderate to excellent yields (Table 2, entries 6 to 18). However, the presence of a strong chelating group (ketone or nitrile) on the arylzinc species inhibits the reaction. As this may be explained by their chelation to the metal center, we attempted those reactions with [$CoBr_2$(bpy)] instead of $CoBr_2$ to form the corresponding ArZnBr and were pleased to observe a reaction. Under these conditions moderate yields of aryl nitriles were obtained (Table 2, entries 19–21), the corresponding ArH was the major product in the reaction mixture. Moreover, despite several attempts, only poor yields of aryl nitrile product were observed from aryl chlorides due to the presence of the excess of pyridine required to prepare the corresponding arylzinc chloride, which hampers the cyanation step.[30] Heteroaryl zinc bromides (thiophene and furyl derivatives) gave only some traces of the expected nitrile derivatives.

In summary, this new cobalt-catalyzed cross-coupling of arylzinc compounds with NCTS allows an easy access to various benzonitriles, and complements nicely known methodologies. This reaction tolerates a large number of functional groups and the yields range from modest to excellent. The presence of a catalytic amount of zinc dust in the coupling reaction is necessary to achieve good efficiency. The low price of the catalyst used, and the mild and 'bench-friendly' conditions for the synthesis of the reagents make this reaction an interesting alternative to more classical methodologies and C−H activation.

Experimental Section

General Procedure for the Synthesis and Cross-Coupling of Arylzinc Bromides with NCTS

To a solution of $CoBr_2$ (0.5 mmol, 110 mg, 13 mol%) and Zn dust (10 mmol, 0.65 g, 2.5 equiv.) in MeCN (4 mL) were added allyl chloride (0.125 mL, 1.5 mmol, 30 mol%) and TFA (0.05 mL), at room temperature under vigorous stirring. This caused a rise in temperature, production of gases, and a change of the color of the mixture from blue to orange to dark grey. Once the orange tinge of the mixture had disappeared, ArBr (3.75 mmol, 1 equiv.) was added. The reaction was followed by GC on iodolyzed aliquots. Once all the starting bromide was consumed, stirring was interrupted; the reaction medium was taken in a 10-mL syringe, and filtered through a syringe filter providing a solution of a cobalt-containing arylzinc bromide solution. An internal standard was added (decane or dodecane), and the solution was titrated by GC on an iodolyzed aliquot. The arylzinc was injected into a solution of NCTS (2.5 mmol) and Zn dust (0.02 g, 0.25 mmol) in acetonitrile which was placed in a Schlenk flask under N_2 and was cooled to 0 °C. The ice bath was allowed to melt, and then the mixture was heated to 50 °C. The reaction was followed by GC on hydrolyzed aliquots, until all of the NCTS was consumed (4–6 h). Aqueous hydrochloric acid (*ca* 2 M, 20 mL) and Et_2O (20 mL) were then added to the reaction mixture and stirred for 5 min. The phases were separated, and the aqueous phase was extracted 3 times with Et_2O (20 mL). The combined organic phases were washed with water (20 mL) and brine (20 mL), dried over $MgSO_4$, and the solvents were evaporated to afford the crude material, which was purified by column chromatography (SiO_2, petroleum ether/diethyl ether) affording the corresponding aryl nitrile compound.

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[30] Noteworthy is that the reaction of arylzinc bromide with NCTS in the presence of pyridine was also less effective than in absence of this additive.

